

tut8

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Q1

```
data = read.csv(file = "https://ckan0.cf.opendata.inter.prod-toronto.ca/dataset/4ef82789-e038-44ef-a478-a8f3590c3eb1/resource/979fb513-5186-41e9-bb23-7b5cc6b89915/download/Pre-2023%20Apartment%20Building%20Evaluations.csv")
```

```
t.test(data$SCORE, mu = 73, alternative="greater")
```

```
##
## One Sample t-test
##
## data: data$SCORE
## t = 8.4873, df = 11759, p-value < 2.2e-16
## alternative hypothesis: true mean is greater than 73
## 95 percent confidence interval:
## 73.66668 Inf
## sample estimates:
## mean of x
## 73.82696
```

```
t.test(data$SCORE, mu = 73, alternative="two.sided")
```

```
##
## One Sample t-test
##
## data: data$SCORE
## t = 8.4873, df = 11759, p-value < 2.2e-16
## alternative hypothesis: true mean is not equal to 73
## 95 percent confidence interval:
## 73.63597 74.01794
## sample estimates:
## mean of x
## 73.82696
```

- ii. The null hypothesis (H_0) in this case is that the population mean (μ) is equal to 73. The alternative hypothesis (H_1) is that the population mean is greater than 73.

The p-value obtained from the t-test is less than the significance level of 0.05. Specifically, the p-value is much smaller, being $< 2.2e-16$ (essentially 0).

Since the p-value is less than 0.05, we reject the null hypothesis. Therefore, there is sufficient evidence to support the researcher's claim that the average score is higher than 73.

Q2

```
sample_prop = mean(data$SCORE > 73)
sample_prop2 = sum(data$SCORE > 73) / length(data$SCORE)
sample_prop
```

```
## [1] 0.5247449
```

```
sample_prop2
```

```
## [1] 0.5247449
```

```
data_2 = data %>% summarize(n = n(), x = sum(SCORE > 73))

data_2
```

```
##           n      x
## 1 11760 6171
```

```
prop.test(data_2$x, data_2$n, p=0.5)
```

```
##
## 1-sample proportions test with continuity correction
##
## data:  data_2$x out of data_2$n, null probability 0.5
## X-squared = 28.704, df = 1, p-value = 8.432e-08
## alternative hypothesis: true p is not equal to 0.5
## 95 percent confidence interval:
##  0.5156700 0.5338035
## sample estimates:
##           p
## 0.5247449
```